Variables and Constants

Variable is a name assign to a storage area that the program can manipulate. A variable type determines the size and layout of the variable's memory.

It also determines the range of values which need to be stored inside that memory and nature of operations that can be applied to that variable.

There are three places where variables you can declare variable programming language:

- Inside a function or a block: Local variables
- Outside of all functions: Global variables
- In the definition of function parameters: Formal parameters

How to declare a variable:
1. Choose the "type" you need.
2. Decide upon a name for the variable.
3. Use the following format for a declaration statement:
   datatype variable identifier;
4. You may declare more than one variable of the same type by separating the variable names with commas.
   int age, weight, height;
5. You can initialize a variable (place a value into the variable location) in a declaration statement.
   double mass = 3.45;

Rules to construct a valid variable name

1. A variable name may consists of letters, digits and the underscore (_) characters.

2. A variable name must begin with a letter. Some system allows to starts the variable name with an underscore as the first character.

3. ANSI standard recognizes a length of 31 characters for a variable name. However, the length should not be normally more than any combination of eight alphabets, digits, and underscores.

4. Uppercase and lowercase are significant. That is the variable **Totamt** is not the same as **totamt** and **TOTAMT.**

5. The variable name may not be a C reserved word (keyword).

Some valid variable names

Total	Amount	ctr	name1
n1	M_age	AMOUNT	

Some invalid variable names

13th	(name)	111	%nm

Naming Conventions

Generally, C programmers maintain the following conventions for naming variables.

- Start a variable name with lowercase letters.
- Try to use meaningful identifiers
- Separate "words" within identifiers with mixed upper and lowercase (for example empCode) or underscores (for example emp_code).
- For symbolic constants use all uppercase letters (for example #define LENGTH 100, #define MRP 45).

The Programming language C has two main variable types

- Local Variables
- Global Variables

Local Variable

- **Local Variable** is defined as a type of variable declared within programming block or subroutines. It can only be used inside the subroutine or code block in which it is declared. The local variable exists until the block of the function is under execution. After that, it will be destroyed automatically.

Example of Local Variable

```
int add()
    {
        int a =4;
        int b=5;
```

```
        return a+b;
    }
```

Here, 'a' and 'b' are local variables

Global Variable

A **Global Variable** in the program is a variable defined outside the subroutine or function. It has a global scope means it holds its value throughout the lifetime of the program. Hence, it can be accessed throughout the program by any function defined within the program, unless it is shadowed.

Example:

```
int a =4;
int b=5;
int add()
{
return a+b;
}
```

Here, 'a' and 'b' are global variables.

Differences between Local and Global Variables

Parameter	Local	Global
Scope	It is declared inside a function.	It is declared outside the function.
Value	If it is not initialized, a garbage value is stored	If it is not initialized zero is stored as default.
Lifetime	It is created when the function starts execution and lost when the functions terminate.	It is created before the program's global execution starts and lost when the program terminates.
Data sharing	Data sharing is not possible as data of the local variable can be accessed by only one function.	Data sharing is possible as multiple functions can access the same global variable.
Parameters	Parameters passing is required for local variables to access the value in other function	Parameters passing is not necessary for a global variable as it is visible throughout the program
Modification of variable value	When the value of the local variable is modified in one function, the changes are not visible in another function.	When the value of the global variable is modified in one function changes are visible in the rest of the program.
Accessed by	Local variables can be accessed with the help of statements, inside a function in which they are declared.	You can access global variables by any statement in the program.
Memory storage	It is stored on the stack unless specified.	It is stored on a fixed location decided by

		the compiler.

Data: Data are characteristics or information, usually numerical, that are collected through observation. In a more technical sense, **data** is a set of values of qualitative or quantitative variables about one or more persons or objects, while a datum (singular of ***data***) is a single value of a single variable.

Information: Information is associated with data, as data represent values attributed to parameters, and information is data in context and with meaning attached. Organized data is called Information.

Bit: A **bit** is a binary digit, the smallest increment of **data** on a computer. The bit represents a logical state with one of two possible values. It's a single unit of information that has a value of either 0 or 1.

Byte: The byte is a unit of digital information that most commonly consists of eight bits.

Constant: A constant can be defined as a fixed value, which is used in algebraic expressions and equations. A constant does not change over time and has a fixed value. For example, the size of a shoe or cloth or any apparel will not change at any point.

In an algebraic expression, x+y = 8, 8 is a constant value, and it cannot be changed.

Difference Between Constants and Variables	
Constant	**Variables**
A constant does not change its value over time.	A variable, on the other hand, changes its value dependent on the equation.
Constants are usually written in numbers.	Variables are specially written in letters or symbols.
Constants usually represent the known values in an equation, expression or in line of programming.	Variables, on the other hand, represent the unknown values.
Constants are used in computer programming.	Variables also have its uses in computer programming and applications.